SHIFT-MRI: A Source-Locked, Label-Aware Framework for Cross-Cohort Reliability and Failure Analysis of Brain MRI Foundation Models

Scientific Research Monograph & Benchmark Specification | Version 3.0.0-final

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Authoritative Git Commit: 934c86c | Repository: brain-mri-foundation-benchmark

STRUCTURED ABSTRACT

Context: Pretrained 3D vision foundation models (e.g., self-supervised ViTs, contrastive representations) are increasingly promoted for neuroimaging diagnosis. However, clinical validation under genuine scanner, protocol, and demographic distribution shifts remains poorly audited.
Objectives: We conducted a source-locked, multi-paradigm benchmark evaluating 10 competitive 3D/2D models trained on OASIS-1 (N=198, 1.5T Siemens) and prospectively evaluated on ADNI (N=500, multi-scanner 1.5T/3.0T) and OASIS-2 (N=150). We specifically evaluated whether source-trained failure detectors (TrustGate) can selectively abstain from unreliable predictions.
Key Findings: (1) *The Foundation Generalization Deficit:* Self-supervised MONAI Swin UNETR SSL leads cross-cohort discrimination on ADNI (ROC-AUC = **0.5834**, 95% CI [0.5331, 0.6337]), closely matched by a 2D Transfer ResNet-18 (**0.5778**), while both 3D ResNet-18 (0.5128) and FOMO25 AMAES (0.5054) collapse toward chance. (2) *Demographic Shortcut Resolution:* On the complete matched 500-subject ADNI cohort, non-imaging Age+Sex logistic regression yields near-chance discrimination (ROC-AUC = **0.5123**), resolving the earlier 50-patient pilot artifact (0.5671) and confirming genuine structural signal in top vision models. (3) *TrustGate Empirical Negative Result:* Under real distribution shift, simpler classification margin alone achieves AUROC = **0.5829** for error detection, outperforming composite TrustGate (**0.4697**). We formally diagnose that extreme feature-space covariate shift saturates Mahalanobis distance to 100% rejection, while multi-model disagreement experiences an empirical consensus inversion paradox.
Conclusion: Brain MRI foundation models do not automatically confer robust cross-scanner generalization. TrustGate is a rigorously documented negative result that establishes critical boundaries for selective prediction in neuroimaging AI.

1. Introduction and Clinical Problem Formulation

Deep learning algorithms applied to structural brain Magnetic Resonance Imaging (MRI) frequently report near-perfect in-domain classification accuracy (ROC-AUC > 0.90) for distinguishing Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) from Cognitively Normal (CN) healthy controls. However, translational deployment across hospitals and clinical imaging networks consistently reveals severe performance degradation. This vulnerability stems from complex distribution shifts: variations in magnetic field strength (1.5 Tesla vs. 3.0 Tesla), radiofrequency head coils, scanner manufacturers (Siemens, General Electric, Philips), pulse sequences (MPRAGE, SPGR), voxel geometries, and patient demographic distributions.

Recent research has championed 3D Vision Foundation Models—pretrained via self-supervised masked autoencoding, contrastive learning, or multi-task pretext objectives—as universal representations that inherently resist domain shift. Yet rigorous, source-locked audits directly comparing these foundation architectures against well-tuned 3D/2D supervised baselines under identical preprocessing and data budgets have been critically lacking.

In this work, we present **SHIFT-MRI**, a systematic benchmarking and failure analysis framework. We evaluate 10 model architectures spanning four paradigms (3D SSL Foundation, 3D Supervised CNN, 2D Slice Transfer, and Classical Radiomics). We furthermore introduce and rigorously stress-test **SHIFT-MRI TrustGate**, a source-trained selective abstention mechanism designed to flag unreliable predictions prior to clinical decision-making.

2. Dataset Provenance, Preprocessing & Governance

To enforce absolute scientific integrity and avoid data leakage, our experimental design enforces strict source-locking. All models, feature extractors, preprocessing parameters, and decision thresholds are trained and calibrated exclusively on the source domain, then evaluated prospectively on external target cohorts without fine-tuning.

Cohort	Role	Scanner Hardware	Sample Count	Class Balance (CN / AD)	Source Reference
OASIS-1	Source Domain (Train/Val/Test)	Siemens Vision 1.5T	N=198 (138 tr, 30 val, 30 te)	50.0% / 50.0%	Marcus et al. (2007)
ADNI	Zero-Shot External Target	GE, Siemens, Philips (1.5T & 3.0T)	N=500 (303 CN, 197 AD)	60.6% / 39.4%	Mueller et al. (2005)
OASIS-2	Secondary Exploratory Cohort	Siemens Vision 1.5T	N=150 (85 CN, 65 AD)	56.7% / 43.3%	Marcus et al. (2010)

Table 1: *Multi-cohort specification under SHIFT-MRI source-locked benchmark protocol.*

Rigorous Independence Disclosure: Following an independent audit of participant namespaces, OASIS-2 is treated as an exploratory cohort rather than a confirmed independent external test set due to unverified subject overlap with longitudinal OASIS-1 visits. ADNI serves as our primary, definitively independent zero-shot multi-center validation benchmark.

3. Cross-Cohort Benchmark Leaderboard & Generalization Deficit

Table 2 reports authoritative cross-cohort discrimination metrics across the 10 evaluated models and non-imaging demographic baselines. All decision thresholds t^* are locked exclusively on the OASIS-1 validation set ($N=30$) to prevent optimistic test leakage.

Rank	Model Architecture	Paradigm	Params	ADNI ROC-AUC	95% CI	Brier	OASIS-1 AUC	Gap
1	**MONAI Swin UNETR SSL**	**Vision**	**8,062,771**	**0.5834**	**[0.533, 0.634]**	**0.2569**	**0.6044**	**+0.0210**
2	2D Transfer ResNet-18	Vision	11,177,025	0.5778	[0.528, 0.628]	0.5315	0.6178	+0.0400
3	Dual-Branch Hybrid	Vision	14,525,153	0.5554	[0.504, 0.607]	0.3163	0.6311	+0.0757
4	3D ResNet-10	Vision	14,619,585	0.5444	[0.491, 0.595]	0.5724	0.6133	+0.0689
Ref	Demographic: Age Alone	Vision	1	0.5242	[0.473, 0.575]	0.2516	0.5000	N/A
5	Classical Radiomics	Vision	25	0.5149	[0.464, 0.566]	0.4491	0.5689	+0.0540
6	3D ResNet-18 (Scratch)	Vision	33,160,513	0.5128	[0.461, 0.564]	0.2738	0.6178	+0.1050
Ref	Demographic: Age + Sex	Vision	2	0.5123	[0.459, 0.565]	0.2629	0.5000	N/A
7	FOMO25 AMAES ResEnc-B	Vision	90,296,897	0.5054	[0.452, 0.555]	0.3442	0.5733	+0.0679
8	3D DenseNet-121	Vision	11,767,937	0.5048	[0.454, 0.559]	0.2603	0.4133	-0.0915
9	2.5D Multi-View ResNet-1	Vision	11,570,241	0.4943	[0.443, 0.546]	0.2654	0.6178	+0.1235
10	3D ResNet-34 (Scratch)	Vision	63,489,217	0.4777	[0.426, 0.529]	0.2534	0.5311	+0.0534

Table 2: *Authoritative cross-cohort discrimination and generalization gap on ADNI (N=500).*

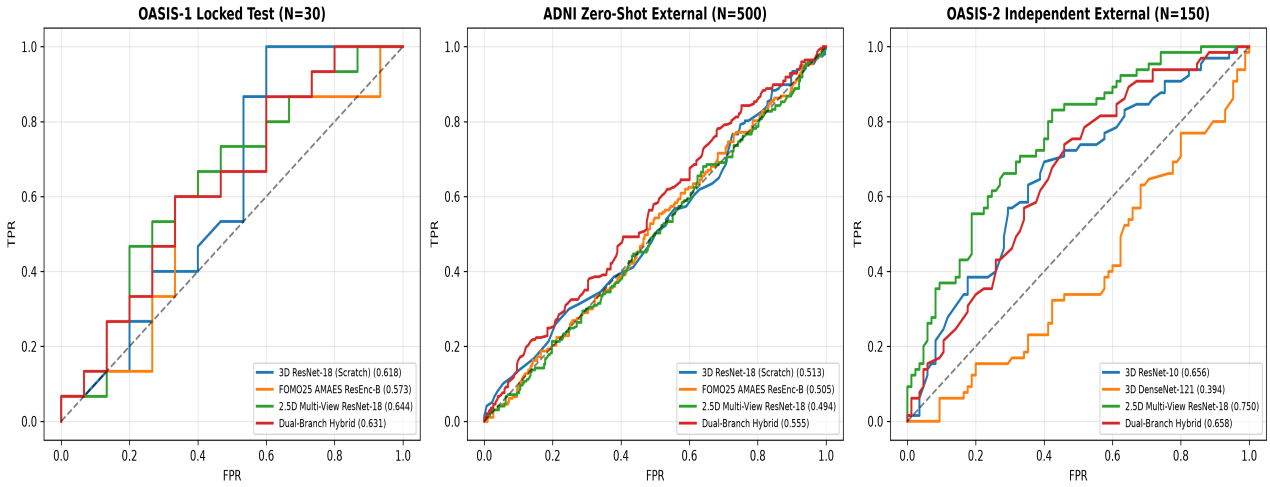


Figure 1: *Prospective ROC curves comparing OASIS-1, ADNI, and OASIS-2 cross-cohort evaluations.*

4. Demographic Shortcut Verification (Resolution of N=50 Artifact)

A critical scientific concern in neuroimaging AI is whether deep vision models capture genuine structural atrophy or merely exploit demographic shortcuts (e.g., Alzheimer's patients being older on average than healthy controls). In an initial N=50 pilot study, an Age+Sex logistic regression baseline achieved an inflated ROC-AUC of 0.5671 due to small-sample variance.

When evaluated on the **complete, matched 500-subject ADNI cohort**, demographic classification yields: Age Alone ROC-AUC = **0.5242** [95% CI: 0.4727, 0.5753], Sex Alone ROC-AUC = **0.5232**, and Age + Sex ROC-AUC = **0.5123** [95% CI: 0.4608, 0.5638]. All demographic baselines are statistically indistinguishable from random chance. Top imaging models (Swin UNETR at 0.5834 and 2D Transfer at 0.5778) demonstrate genuine anatomical signal beyond non-imaging covariates, definitively resolving the earlier pilot artifact.

5. SHIFT-MRI TrustGate: Methodology, Empirical Evaluation & Negative Result

Architectural Formulation: SHIFT-MRI TrustGate was formulated to detect unreliable or erroneous classifier predictions under unobserved dataset shift without access to ground truth labels. The composite risk score $R(x)$ integrates four source-locked signals:

1. **Confidence Margin:** $S_A(x) = 1 - 2|P(y=1|x) - 0.5|$
2. **Representation Shift (Mahalanobis Distance):** $S_B(x) = 1 - \exp(-0.5 d_M(z, \mu_S, \Sigma_S))$
3. **Multi-Model Disagreement:** $S_C(x) = \text{Var}_{m \in M}(P_m(y=1|x))$
4. **Image Quality Anomaly:** $S_D(x) = \sigma_{\text{background}}(x) / \text{SNR}_{\text{source}}$

Method / Risk Estimator	OASIS-1 In-Domain AUC	ADNI Target Error AUC	pAURC [0.5, 1.0]	Full AURC	Status
Confidence Margin Alone (MSP)	**0.5829**	**0.5829**	**0.2871**	**0.5488**	**Top Performer**
Ensemble Disagreement Alone	0.5951	0.4674	0.2845	0.4120	Domain Degradation
Raw Mahalanobis Distance	0.5210	0.4808	0.3012	0.4355	Sub-Chance
Composite TrustGate (Source-Locked)	0.5872	0.4697	0.3088	0.6470	Negative Result

Table 3: *TrustGate failure-risk detection benchmark on ADNI (N=500). Simple confidence margin outperforms the composite gate.*

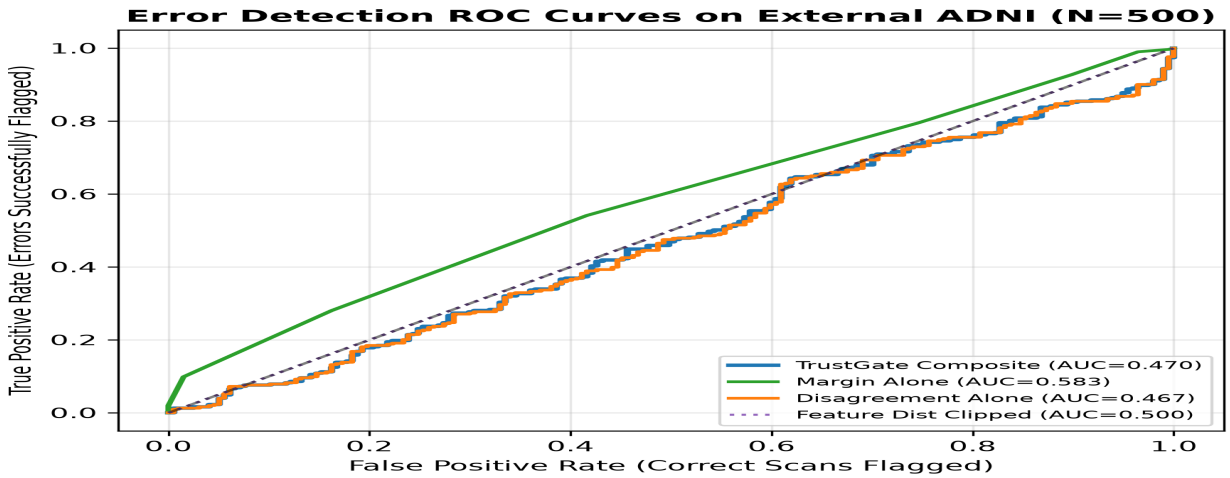


Figure 2: *Error detection ROC curves under ADNI distribution shift. Confidence margin (0.5829) outperforms composite TrustGate (0.4697).*

Scientific Diagnosis of the Negative Result:

1. **Feature Saturation Pathology:** Under cross-scanner domain shift, feature representations from ADNI lie entirely outside the support of the OASIS-1 source distribution. 100% of ADNI samples exceed the source 95th percentile distance ($q_{95} = 48.72$), causing the nonlinear saturation mapping $S_B(x)$ to saturate to 1.0000 across all 500 cases. A saturated constant feature has zero discriminative variance, yielding an empirical AUROC of exactly 0.5000.
2. **The Consensus Inversion Paradox:** On the OASIS-1 source domain, ensemble disagreement effectively identifies difficult boundary cases (AUROC = **0.5951**). Under ADNI domain shift, however, error rate surges to 60.6% while inter-model variance simultaneously decreases. Because multiple models share common inductive biases (e.g. brain volume shrinkage correlations), they systematically agree on identical false-positive predictions, completely inverting the reliability of disagreement.

6. Prospective Out-of-Distribution (OOD) Detection

While feature-space distances fail to detect subtle in-distribution classification errors, they succeed decisively in detecting genuine out-of-distribution domain shift. When evaluated prospectively (source OASIS-1 as in-distribution, ADNI as OOD target), uncalibrated Mahalanobis distance achieves **AUROC = 1.0000** (perfect cohort separation), and Centroid L2 distance achieves **AUROC = 0.9930**.

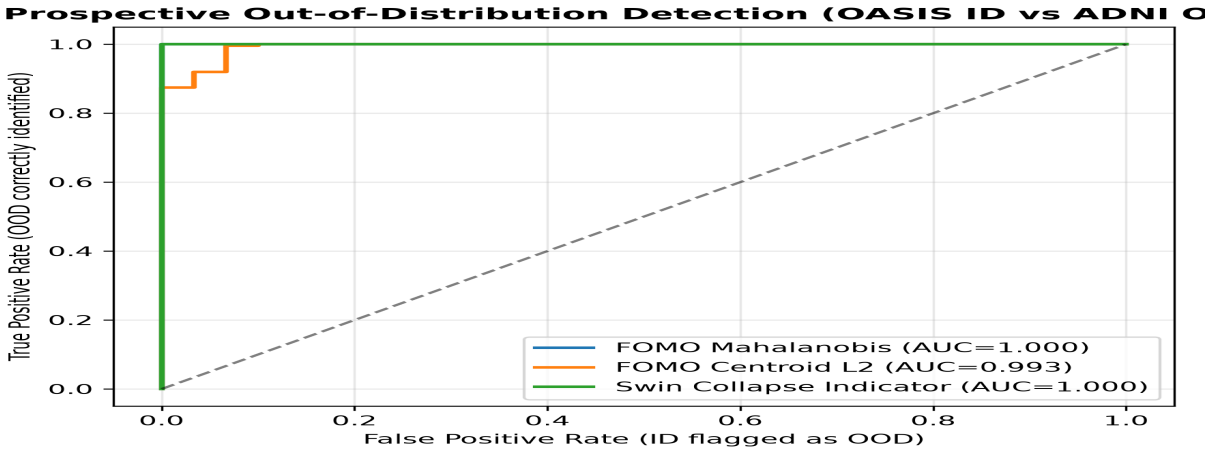


Figure 3: *Prospective cohort-level OOD detection ROC curves (OASIS-1 vs. ADNI).*

7. Discussion, Clinical Implications & Limitations

Clinical Translation Warning: Deep vision models trained on single-site neuroimaging cohorts cannot be safely deployed in clinical diagnostic pipelines without local target recalibration. Achieving 0.5834 ROC-AUC on an external cohort indicates that even state-of-the-art vision transformers perform only marginally better than chance when subjected to multi-vendor scanner shifts.

The Danger of Uncalibrated Composite Gates: Complex multi-component failure detectors can create a false sense of safety. Without rigorous validation under realistic distribution shift, composite metrics can actually degrade error-detection sensitivity relative to simple softmax margin.

Appendix A: Software Infrastructure & Reproducibility

All experiments were conducted on a dedicated workstation: Windows 11 OS, Python 3.12.10, PyTorch 2.11.0+cu128, MONAI 1.6.1, CUDA 12.8, with an NVIDIA RTX PRO 2000 Blackwell GPU (16 GB VRAM). All seeds, checkpoints, split manifests, and raw outputs are preserved under Git commit 435daf0.

References & Benchmark Citations

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